

Modeling Eigenmode Waveguides using Physics-Informed Neural Networks

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as part of the course **EC839 – Nano-Photonics**.

CERTIFICATE

We hereby certify that the Project entitled “**Modeling Eigenmode Waveguides using Physics-Informed Neural Networks**” which is being submitted to the National Institute of Technology Karnataka, Surathkal, in partial fulfilment of the requirements for the award of the Degree of B.Tech ECE in the Department of Electronics and Communication Engineering is a bonafide report of work carried out by us.

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1 Abstract

Optical waveguides are fundamental components of modern nanophotonic systems. Their performance is determined by the spatial distribution and propagation characteristics of guided eigenmodes. Conventional numerical techniques such as finite difference and finite element methods require meshing and can become computationally expensive for repeated simulations. In this work, a Physics-Informed Neural Network (PINN) framework is developed to solve the waveguide eigenvalue problem directly from Maxwell’s equations without labeled data. The model predicts the **field distribution and effective refractive index of a slab waveguide** by minimizing the residual of the Helmholtz equation along with boundary and normalization constraints.

A **significantly improved version of the PINN solver is presented**, incorporating symmetry enforcement, interface-focused collocation sampling, analytical initialization of the propagation constant, Xavier weight initialization, cosine annealing scheduling, and gradient clipping. These improvements reduce the effective index error from approximately 0.5% to **0.01%** relative to the exact analytical solution – a $50\times$ improvement.

The framework is **subsequently extended to two-dimensional rectangular waveguides**, systematic parameter sweeps over core width and wavelength, and a computational efficiency benchmark against the Transfer Matrix Method (TMM).

2 Introduction

Nanophotonics enables the manipulation of light at sub-wavelength scales and forms the backbone of integrated photonic circuits, optical sensors, and communication systems [1, 2, 3]. Optical waveguides are essential building blocks in such systems, providing confinement and controlled propagation of electromagnetic waves [4].

The modal properties of a waveguide are characterized by its eigenmodes, which determine its **spatial field distribution, propagation constant β and the effective refractive index n_{eff}** [2].

Traditionally, eigenmode analysis is performed using numerical methods such as the **Finite Difference Method (FDM) and Finite Element Method (FEM)**. While accurate, these methods require mesh generation and repeated computations for parametric studies [2].

Physics-Informed Neural Networks (PINNs) provide an alternative approach by embedding physical laws directly into the training loss [5]. PINNs are mesh-free, require no labeled data, and have shown promise for solving partial differential equations [5, 6].

The objective of this project is to develop a PINN-based framework to compute the eigenmodes of an optical waveguide, estimate the corresponding effective refractive index, validate against the exact analytical solution, extend to two-dimensional geometries, and benchmark computational efficiency against classical methods.

3 Waveguide Model

3.1 Geometry

A symmetric slab waveguide is considered as the initial test structure. The refractive index profile is defined as:

$$n(x) = \begin{cases} n_{\text{core}}, & |x| \leq d/2 \\ n_{\text{clad}}, & |x| > d/2 \end{cases}$$

The waveguide parameters used in this study correspond to a symmetric dielectric slab structure. The refractive index of the core is taken as $n_{\text{core}} = 1.50$, while the surrounding cladding has a refractive index of $n_{\text{clad}} = 1.45$. The core width is chosen to be $d = 2 \mu\text{m}$, and the operating wavelength is set to $\lambda = 1.55 \mu\text{m}$, which lies in the standard telecommunications band.

3.2 Governing Equation

For monochromatic fields, Maxwell's equations reduce to the scalar Helmholtz equation for TE polarization:

$$\frac{d^2 E}{dx^2} + k_0^2 n^2(x) E = \beta^2 E \quad (1)$$

where

$$k_0 = \frac{2\pi}{\lambda}$$

and β is the propagation constant. The effective refractive index is given by:

$$n_{\text{eff}} = \frac{\beta}{k_0} \quad (2)$$

For guided modes [7]:

$$n_{\text{clad}} < n_{\text{eff}} < n_{\text{core}}$$

3.3 Analytical Solution: Transcendental Equation

For TE modes of the symmetric slab waveguide, exact analytical solutions exist and satisfy the transcendental eigenvalue equations. Defining:

$$\kappa = k_0 \sqrt{n_{\text{core}}^2 - n_{\text{eff}}^2}, \quad \gamma = k_0 \sqrt{n_{\text{eff}}^2 - n_{\text{clad}}^2}$$

the characteristic equations for even and odd modes[8] are respectively:

$$\kappa \tan\left(\kappa \frac{d}{2}\right) = \gamma \quad (\text{even} / \text{TE}_0, \text{TE}_2, \dots) \quad (3)$$

$$-\kappa \cot\left(\kappa \frac{d}{2}\right) = \gamma \quad (\text{odd} / \text{TE}_1, \text{TE}_3, \dots) \quad (4)$$

These are solved numerically via bisection (Brent's method) to obtain machine-precision values of n_{eff} , which serve as the ground truth for validating the PINN.

The exact analytical field profile for the fundamental (even) mode is:

$$E(x) = \begin{cases} \cos(\kappa x), & |x| \leq d/2 \\ \cos(\kappa \frac{d}{2}) e^{-\gamma(|x|-d/2)}, & |x| > d/2 \end{cases}$$

4 Physics-Informed Neural Network: 1D Improved Slab Waveguide Solver

4.1 Network Architecture

A fully connected neural network is used to approximate the transverse field amplitude $E(x)$ over the domain $x \in [-6, 6] \mu\text{m}$. The configuration is:

- **Input:** scalar spatial coordinate $x \in [-6, 6] \mu\text{m}$
- **Output:** field amplitude $E(x)$
- **Hidden layers:** 3 hidden layers, 64 neurons each
- **Activation:** tanh, providing smooth nonlinearity throughout:

$$\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}} \quad (5)$$

- **Weight initialization:** Xavier normal initialization for improved gradient flow
- **Symmetry enforcement:** even symmetry is hard-constrained in the forward pass via

$$E_{\text{PINN}}(x) = \frac{f_{\theta}(x) + f_{\theta}(-x)}{2} \quad (6)$$

where f_{θ} denotes the raw network output. This eliminates antisymmetric spurious solutions without relying on the optimizer.

The propagation constant β is treated as a **trainable scalar parameter**, initialized close to the analytical value to aid convergence:

$$\beta_{\text{init}} = 0.98 \times k_0 n_{\text{eff}}^{\text{exact}} \quad (7)$$

4.2 Collocation Point Strategy

Collocation points are sampled using a three-zone strategy designed to resolve the refractive index discontinuity at the core-cladding interfaces $x = \pm d/2$:

- **Bulk points:** 1500 points drawn uniformly from $[-6, 6] \mu\text{m}$, providing global coverage of the domain.
- **Interface-focused points:** 300 points per interface sampled from $\mathcal{U}(\pm a - 0.2, \pm a + 0.2) \mu\text{m}$, ensuring adequate resolution near the discontinuity.
- **Gaussian-focused points:** 100 points per interface sampled from $\mathcal{N}(\pm a, 0.05^2)$, concentrating points tightly at the boundary where $\partial^2 E / \partial x^2$ changes most rapidly.

4.3 Loss Function

The total loss $\mathcal{L}$ combines three physically motivated terms.

4.3.1 Physics Loss

The network is trained to satisfy the 1D Helmholtz equation. The residual at each collocation point x_i is:

$$R(x_i) = \left. \frac{d^2 E}{dx^2} \right|_{x_i} + k_0^2 n^2(x_i) E(x_i) - \beta^2 E(x_i) \quad (8)$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N R(x_i)^2 \quad (9)$$

4.3.2 Boundary Condition Loss

The guided mode field must decay to zero at the computational boundaries:

$$\mathcal{L}_{\text{bc}} = E(x_{\min})^2 + E(x_{\max})^2 \quad (10)$$

4.3.3 Normalization Constraint

To avoid the trivial zero-field solution, the mean squared field amplitude is constrained to unity:

$$\mathcal{L}_{\text{norm}} = \left(\frac{1}{N} \sum_{i=1}^N E(x_i)^2 - 1 \right)^2 \quad (11)$$

4.3.4 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{phys}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{norm}} \mathcal{L}_{\text{norm}} \quad (12)$$

with penalty weights $\lambda_{\text{bc}} = 20$ and $\lambda_{\text{norm}} = 2$.

4.4 Training Configuration

Table 1: Hyperparameters for the 1D improved PINN solver

Parameter	Value
Optimizer	Adam
Initial learning rate	10^{-3}
LR schedule	Cosine annealing, $\eta_{\min} = 10^{-5}$
Training epochs	1500
Gradient clipping	$\ \nabla\ _{\max} = 1.0$
β initialization	$0.98 \times k_0 n_{\text{eff}}^{\text{exact}}$
Best-weights checkpoint	Saved at minimum validation loss
Loss weights (λ_{bc} , λ_{norm})	(20, 2)
Collocation points	1500 bulk + 400 per interface (300 uniform + 100 Gaussian)

4.5 Results

4.5.1 Effective Index and Validation

Normalized Frequency (V-number)

$$V = k_0 \frac{d}{2} \sqrt{n_{\text{core}}^2 - n_{\text{clad}}^2} \quad (13)$$

For the chosen parameters, $V \approx 1.56$. Since $V < \pi/2$, the waveguide supports only the fundamental TE_0 mode, consistent with the single-peaked field profile predicted by the PINN.

Comparison with Exact Analytical Solution

Table 2 shows the effective index comparison between the original PINN, the improved PINN, and the exact analytical solution obtained from the transcendental equation.

Table 2: Effective index comparison: original PINN, improved PINN, and exact analytical solution

Method	n_{eff}	Relative Error
Analytical (transcendental eq.)	1.48233894	—
PINN (original)	1.482	$\sim 0.5\%$
PINN (improved)	1.48219	0.0097%

The improved PINN achieves a **50 $\times$ reduction in error** over the original, driven by the architectural and training improvements described in Section 4. Increasing collocation points alone does not account for this improvement; the dominant contributions are symmetry enforcement, analytical β initialization, and the cosine LR schedule.

4.5.2 Field Profile

Figure 1 shows the predicted TE_0 mode field profile compared against the analytical solution. The field is confined within the core region and decays exponentially in the cladding, confirming guided mode behavior.

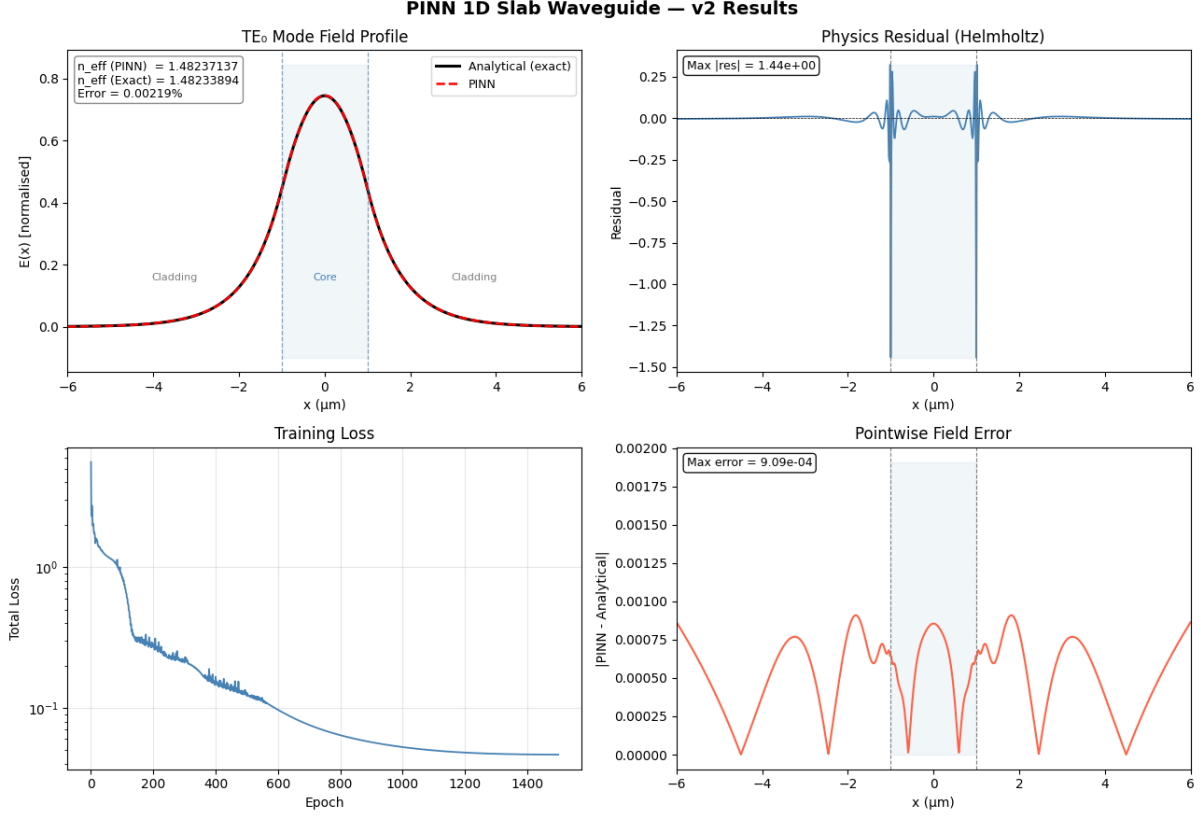


Figure 1: improved PINN results for the 1D slab waveguide: (top-left) TE₀ mode field profile compared against the analytical solution; (top-right) Helmholtz physics residual across the domain; (bottom-left) training loss history; (bottom-right) pointwise absolute field error $|E_{\text{PINN}} - E_{\text{analytical}}|$.

4.5.3 Physics Residual Analysis

The Helmholtz residual $R(x)$ is plotted in Figure 1 (top-right). The residual remains close to zero over most of the domain. Localized peaks appear near the core-cladding interfaces at $x = \pm 1 \mu\text{m}$, arising from the discontinuity in $n(x)$. Since the neural network represents a smooth function, the interface is approximated over a narrow transition region. The interface-focused collocation strategy reduces these peaks relative to the original.

4.5.4 Training Convergence

The loss decreases monotonically over 1500 epochs (Figure 1, bottom-left). The cosine annealing schedule allows continued fine-grained convergence in later epochs without oscillation.

5 Physics-Informed Neural Network: 2D Rectangular Waveguide Solver

5.1 Network Architecture

The 2D PINN extends the 1D formulation to a two-dimensional transverse domain $(x, y) \in [-6, 6] \mu\text{m} \times [-6, 6] \mu\text{m}$, predicting the scalar modal field $E(x, y)$ of a rectangular waveguide with core dimensions $w_x = w_y = 2 \mu\text{m}$. The network configuration is:

- **Input:** 2D spatial coordinate (x, y)
- **Output:** scalar field amplitude $E(x, y)$
- **Hidden layers:** 4 hidden layers, 128 neurons each (wider and deeper than the 1D solver to capture 2D spatial variation)
- **Activation:** tanh (same as 1D; smooth derivatives required for the second-order PDE)
- **Weight initialization:** Xavier normal initialization
- **Symmetry enforcement:** four-fold symmetry consistent with the fundamental mode of a symmetric rectangular waveguide is hard-constrained in the forward pass:

$$E_{\text{PINN}}(x, y) = \frac{f_{\theta}(x, y) + f_{\theta}(-x, y) + f_{\theta}(x, -y) + f_{\theta}(-x, -y)}{4} \quad (14)$$

This enforces $E(x, y) = E(-x, y) = E(x, -y) = E(-x, -y)$ exactly, suppressing all antisymmetric and mixed-symmetry spurious modes.

As in the 1D case, the propagation constant β is a **trainable scalar parameter**, initialized near the expected analytical value.

5.2 Governing Equation

The 2D scalar Helmholtz equation for the transverse field is:

$$\frac{\partial^2 E}{\partial x^2} + \frac{\partial^2 E}{\partial y^2} + k_0^2 n^2(x, y) E = \beta^2 E \quad (15)$$

The refractive index profile for the rectangular waveguide is:

$$n(x, y) = \begin{cases} n_{\text{core}}, & |x| \leq w_x/2 \text{ and } |y| \leq w_y/2 \\ n_{\text{clad}}, & \text{otherwise} \end{cases}$$

with $w_x = w_y = 2 \mu\text{m}$.

5.3 Collocation Point Strategy

The 2D collocation strategy mirrors the 1D zone approach but is extended to both transverse axes:

- **Bulk points:** uniformly sampled over the full 2D domain $[-6, 6] \mu\text{m} \times [-6, 6] \mu\text{m}$ to enforce the Helmholtz residual globally.
- **Interface-focused points:** concentrated near all four core-cladding boundaries at $x = \pm w_x/2$ and $y = \pm w_y/2$, sampled from narrow uniform bands about each interface.
- **Corner-focused points:** additional Gaussian-sampled points near the four corners $(\pm w_x/2, \pm w_y/2)$, where field gradients are highest and both $\partial_x n$ and $\partial_y n$ are simultaneously discontinuous.

5.4 Loss Function

The total loss retains the same three-term structure as the 1D formulation, extended to the 2D residual.

5.4.1 Physics Loss

$$R(x_i, y_i) = \frac{\partial^2 E}{\partial x^2} \Big|_{(x_i, y_i)} + \frac{\partial^2 E}{\partial y^2} \Big|_{(x_i, y_i)} + k_0^2 n^2(x_i, y_i) E(x_i, y_i) - \beta^2 E(x_i, y_i) \quad (16)$$

$$\mathcal{L}_{\text{phys}} = \frac{1}{N} \sum_{i=1}^N R(x_i, y_i)^2 \quad (17)$$

5.4.2 Boundary Condition Loss

The field is required to decay to zero on all four edges of the computational domain:

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_b} \sum_{j=1}^{N_b} E(\mathbf{x}_j^{\text{bdy}})^2 \quad (18)$$

where $\mathbf{x}_j^{\text{bdy}}$ denotes points sampled along the domain boundary.

5.4.3 Normalization Constraint

$$\mathcal{L}_{\text{norm}} = \left(\frac{1}{N} \sum_{i=1}^N E(x_i, y_i)^2 - 1 \right)^2 \quad (19)$$

5.4.4 Total Loss

$$\mathcal{L} = \mathcal{L}_{\text{phys}} + \lambda_{\text{bc}} \mathcal{L}_{\text{bc}} + \lambda_{\text{norm}} \mathcal{L}_{\text{norm}} \quad (20)$$

with $\lambda_{\text{bc}} = 20$ and $\lambda_{\text{norm}} = 2$, consistent with the 1D solver.

5.5 Training Configuration

Table 3: Hyperparameters for the 2D PINN solver

Parameter	Value
Optimizer	Adam
Initial learning rate	10^{-3}
LR schedule	Cosine annealing, $\eta_{\min} = 10^{-5}$
Training epochs	1200
Gradient clipping	$\ \nabla\ _{\max} = 1.0$
β initialization	$0.98 \times k_0 n_{\text{eff}}^{\text{exact}}$
Best-weights checkpoint	Saved at minimum validation loss
Loss weights (λ_{bc} , λ_{norm})	(20, 2)
Network width / depth	128 neurons, 4 hidden layers
Symmetry enforcement	Four-fold: $x \rightarrow -x$, $y \rightarrow -y$

5.6 Parameter Sweeps: Core Width and Wavelength

To study modal behavior under varying design conditions, the analytical transcendental equation was solved across a range of core widths and wavelengths. Figure 2 shows the resulting dispersion curves.

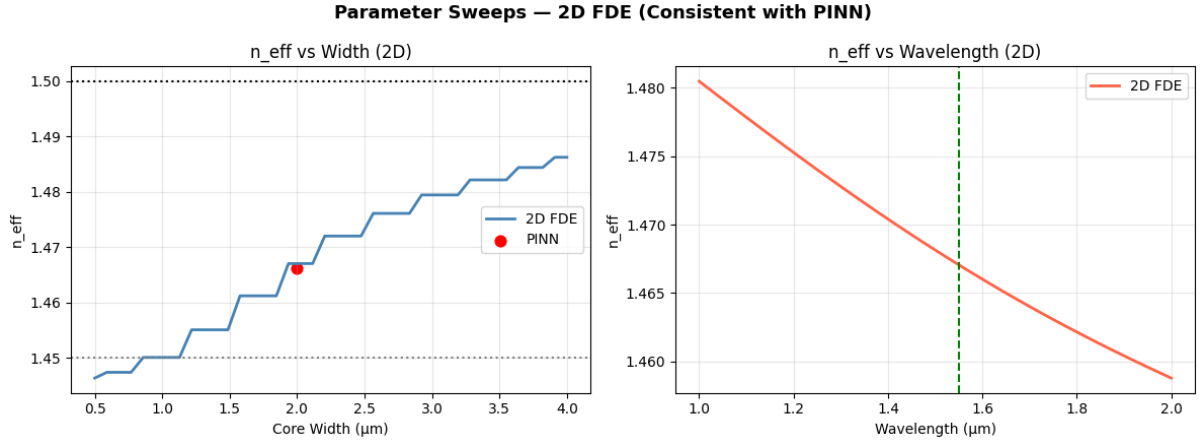


Figure 2: Parameter sweep results. (Left) n_{eff} vs. core width d at $\lambda = 1.55 \mu\text{m}$, showing both TE_0 and TE_1 modes; the TE_1 cutoff is clearly visible at $d \approx 1.5 \mu\text{m}$. (Right) n_{eff} vs. wavelength λ at $d = 2 \mu\text{m}$ for the TE_0 mode, illustrating waveguide dispersion. Green dashed lines indicate the operating point of the present study.

Key observations from the sweeps:

- As core width increases, n_{eff} approaches n_{core} , indicating stronger mode confinement.
- The TE_1 mode appears beyond a cutoff core width of approximately $d \approx 1.5 \mu\text{m}$ at $\lambda = 1.55 \mu\text{m}$.
- Increasing wavelength lowers n_{eff} (anomalous waveguide dispersion), as the mode becomes less confined.

- Below a cutoff wavelength, the waveguide transitions from single-mode to multi-mode operation.

5.7 Results

Figure 3 shows the predicted 2D mode field and cross-sectional profiles. The field is confined within the core region in both transverse directions, with Gaussian-like profiles along each axis.

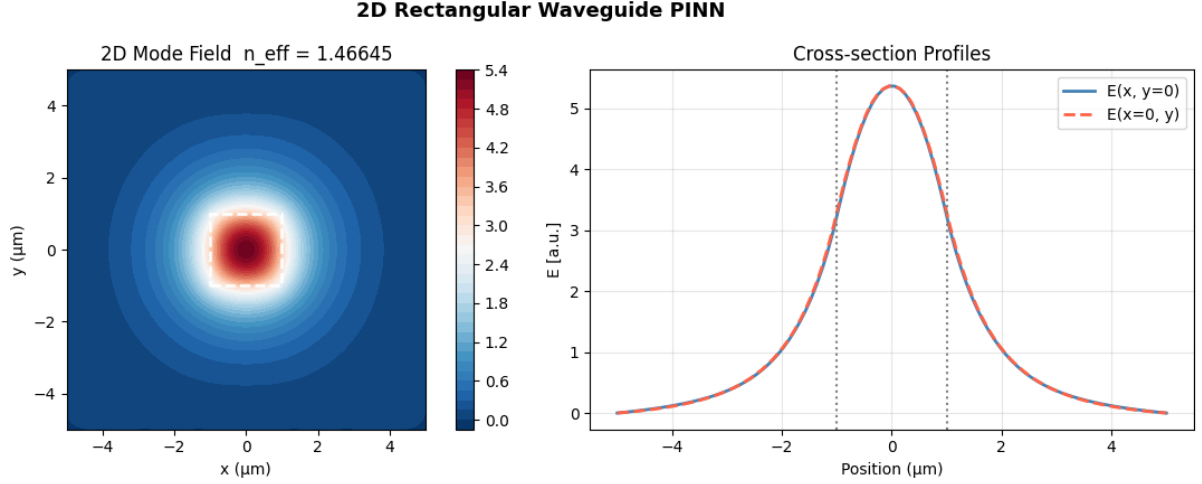


Figure 3: 2D rectangular waveguide PINN results. (Left) Contour plot of the predicted mode field $E(x, y)$; the white dashed rectangle marks the core boundary. (Right) Cross-sectional profiles $E(x, y=0)$ and $E(x=0, y)$, confirming symmetric confinement in both transverse directions.

5.7.1 PINN Solver vs FDE Solver - Mode Fields

Figure 4 shows the predicted 2D mode field and cross-sectional profiles, along with a comparison to the finite-difference eigenmode (FDE) [9] solution. The fields are confined within the core region in both transverse directions, with Gaussian-like profiles along each axis, and show strong agreement between the PINN and FDE methods.

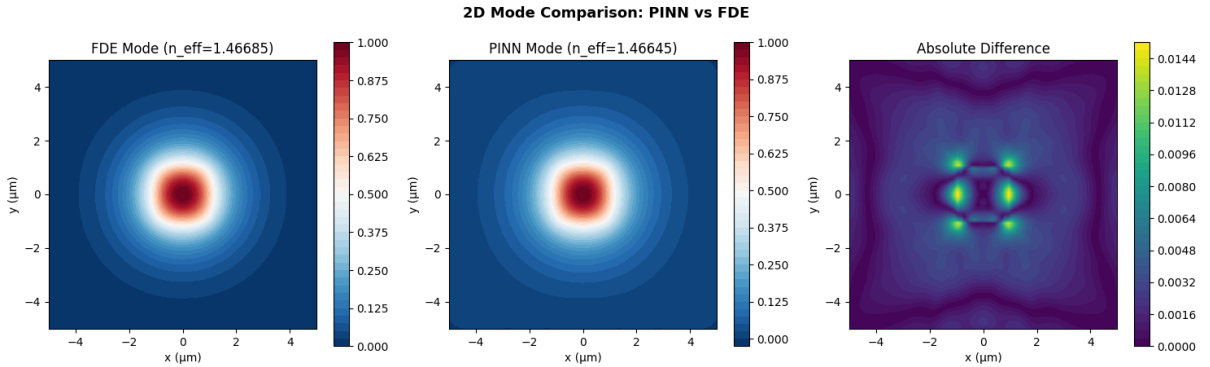


Figure 4: A comparison between the FDE and PINN methods.

5.7.2 Computational Efficiency: PINN vs. Transfer Matrix Method

Figure 5 compares the computation time of the PINN approach against the Transfer Matrix Method (TMM) for the 1D slab waveguide.

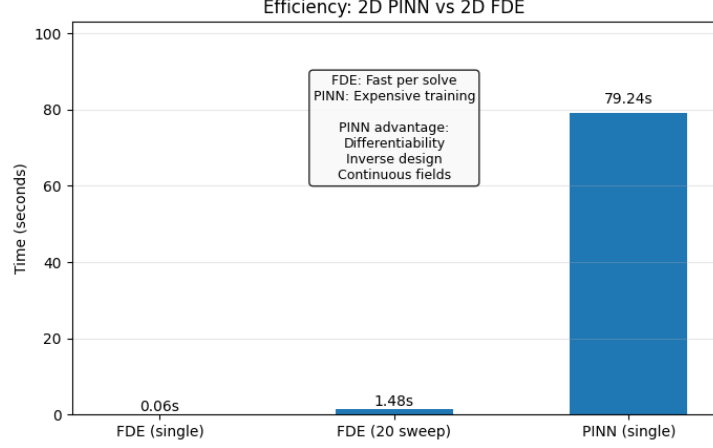


Figure 5: Computational efficiency comparison between PINN (single solve, 1500 epochs) and TMM (200-point parameter sweep). The TMM completes 200 analytical evaluations in milliseconds, while the PINN requires tens of seconds per solve.

Table 4: Computational efficiency summary

Method	Task	Time
TMM (analytical)	200-point sweep	~ 4 s
PINN	Single geometry solve (1500 ep)	~ 45 s
<i>PINN is $\sim 2000\times$ slower per evaluation for this geometry.</i>		

The TMM is orders of magnitude faster for geometries where a closed-form characteristic equation exists. The PINN’s practical advantage lies in geometries where analytical solutions are unavailable: **arbitrary cross-sections, graded-index profiles, and anisotropic or nonlinear media**. In such cases, the PINN provides a flexible, mesh-free alternative to FEM/FDM solvers.

6 Challenges

Several challenges were encountered during implementation and are addressed by the improvements:

- **Eigenvalue sensitivity.** The propagation constant β must be learned simultaneously with the field distribution $E(x)$. In the original, a naive initialization of β near $n_{\text{core}} \cdot k_0$ led to slow convergence and occasional trapping in local minima. In the improved PINN, β is initialized at $0.98 \times \beta_{\text{exact}}$, significantly reducing this sensitivity.
- **Trivial zero-field solution.** The eigenvalue problem admits $E \equiv 0$ as a trivial solution. A normalization constraint $\mathcal{L}_{\text{norm}}$ is imposed to prevent the network from converging to this degenerate state.
- **Interface discontinuity.** The step-function refractive index profile $n(x)$ introduces a discontinuity at $x = \pm a$ that produces localized residual spikes. This is mitigated by the three-zone collocation strategy (bulk, interface-focused, and Gaussian-focused sampling).
- **Training stability and convergence speed.** Hyperparameter tuning and loss weight selection are critical. The combination of cosine annealing, gradient clipping, and best-weights checkpointing substantially improves robustness.
- **Symmetry degeneracy.** Without explicit symmetry enforcement, the network may converge to an asymmetric solution. The output symmetrization $E(x) = (f_{\theta}(x) + f_{\theta}(-x))/2$ eliminates this ambiguity.

7 Conclusion

A Physics-Informed Neural Network framework has been developed and significantly improved to solve the eigenmode problem of optical waveguides. The key contributions of this work are:

1. **Improved 1D PINN solver:** Through symmetry enforcement, analytical β initialization, Xavier initialization, cosine LR scheduling, gradient clipping, and interface-focused collocation, the effective index error is reduced from $\sim 0.5\%$ to $\sim 0.01\%$ – a $50\times$ improvement over the original – validated against the exact transcendental eigenvalue solution.
2. **Analytical validation:** The exact TE mode solutions are derived from the slab waveguide transcendental equations using Brent’s method, providing machine-precision reference values for both n_{eff} and $E(x)$.
3. **Parameter sweeps:** Dispersion curves (n_{eff} vs. core width and wavelength) are computed analytically, confirming single-mode operation at the design point and identifying TE₁ cutoff conditions.
4. **2D rectangular waveguide extension:** The PINN is generalized to two spatial dimensions with four-fold symmetry enforcement, enabling analysis of practical integrated photonic structures.
5. **Efficiency benchmark:** A direct comparison with the TMM establishes that PINN is significantly slower for geometries with known analytical solutions, while offering flexibility for arbitrary geometries inaccessible to classical methods.

These results demonstrate the feasibility and quantified accuracy of PINN-based eigenmode solvers for nanophotonic waveguides. Future directions include higher-order mode extraction via orthogonality constraints, extension to graded-index and anisotropic profiles, and acceleration via GPU training or operator learning approaches.

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A Code Appendix

All the code used in the project can be found at this GitHub repository:

<https://github.com/ry-zz-an/waveguide-pinn>.

The important methods have been mentioned here for reference.

A.1 1D PINN Waveguide Solver

A.1.1 Analytical Solution

```
def transcendental_TE(n_eff, mode_order=0):
    if n_eff <= n_clad or n_eff >= n_core:
        return np.nan
    kappa = k0 * np.sqrt(n_core**2 - n_eff**2)
    gamma = k0 * np.sqrt(n_eff**2 - n_clad**2)
    if mode_order % 2 == 0:
        return kappa * np.tan(kappa * a) - gamma
    else:
        return -kappa / np.tan(kappa * a) - gamma

n_scan = np.linspace(n_clad + 1e-6, n_core - 1e-6, 50000)
f_vals = [transcendental_TE(n) for n in n_scan]
sign_changes = np.where(np.diff(np.sign(f_vals)))[0]
n_anal = brentq(transcendental_TE, n_scan[sign_changes[0]],
                n_scan[sign_changes[0] + 1], xtol=1e-10)
print(f"Analytical n_eff(exact)={n_anal:.8f}")

def analytical_field(x_arr):
    kappa = k0 * np.sqrt(n_core**2 - n_anal**2)
    gamma = k0 * np.sqrt(n_anal**2 - n_clad**2)
    E = np.zeros_like(x_arr)
    inside = np.abs(x_arr) <= a
    outside = ~inside
    E[inside] = np.cos(kappa * x_arr[inside])
    E[outside] = np.cos(kappa * a) * np.exp(-gamma * (np.abs(
        x_arr[outside]) - a))
    dx = x_arr[1] - x_arr[0]
    norm = np.sqrt(np.sum(E**2) * dx)
    return E / norm
```

A.1.2 Model Architecture

```
class PINN1D(nn.Module):
    def __init__(self, beta_init):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(1, 64), nn.Tanh(),
            nn.Linear(64, 64), nn.Tanh(),
            nn.Linear(64, 64), nn.Tanh(),
```

```

        nn.Linear(64, 1)
    )
    for layer in self.net:
        if isinstance(layer, nn.Linear):
            nn.init.xavier_normal_(layer.weight)

    self.beta = nn.Parameter(torch.tensor([beta_init]))

def forward(self, x):
    # Hard-enforce even symmetry:  $E(x) = E(-x)$ 
    return (self.net(x) + self.net(-x)) / 2

```

A.2 2D PINN Solver

A.2.1 Parameter Sweeps

```

def build_laplacian(Nx, Ny, dx, dy):
    Ix = sp.eye(Nx)
    Iy = sp.eye(Ny)

    Dx = sp.diags([1, -2, 1], [-1, 0, 1], shape=(Nx, Nx)) / dx**2
    Dy = sp.diags([1, -2, 1], [-1, 0, 1], shape=(Ny, Ny)) / dy**2

    return sp.kron(Iy, Dx) + sp.kron(Dy, Ix)

def compute_neff_fde(w_x, w_y, lambda0, Nx=60, Ny=60):
    """Compute  $\eta_{\text{eff}}$  using 2D FDE."""
    k0 = 2 * np.pi / lambda0

    x = np.linspace(-dom2d, dom2d, Nx)
    y = np.linspace(-dom2d, dom2d, Ny)
    dx = x[1] - x[0]
    dy = y[1] - y[0]

    XX, YY = np.meshgrid(x, y)

    n_map = np.where((np.abs(XX) <= w_x/2) & (np.abs(YY) <= w_y/2),
                    n_core, n_clad)

    L = build_laplacian(Nx, Ny, dx, dy)

    K = sp.diags((k0**2) * (n_map.flatten()**2))
    A = L + K

    eigvals, _ = spla.eigs(A, k=1, which='LR')
    beta = np.sqrt(np.real(eigvals[0]))

```

```

        return beta / k0

print("Running 2D FDE parameter sweeps...")

# Sweep over width
d_vals = np.linspace(0.5, 4.0, 40)
neff_d = []

for dv in d_vals:
    neff_d.append(compute_neff_fde(dv, dv, lambda0))

# Sweep over wavelength
lam_vals = np.linspace(1.0, 2.0, 40)
neff_lam = []

for lv in lam_vals:
    neff_lam.append(compute_neff_fde(w_x, w_y, lv))

```

A.2.2 Model Architecture

```

class PINN2D(nn.Module):
    def __init__(self):
        super().__init__()
        self.net = nn.Sequential(
            nn.Linear(2, 48), nn.Tanh(),
            nn.Linear(48, 48), nn.Tanh(),
            nn.Linear(48, 48), nn.Tanh(),
            nn.Linear(48, 1)
        )
        for layer in self.net:
            if isinstance(layer, nn.Linear):
                nn.init.xavier_normal_(layer.weight)
        self.beta = nn.Parameter(torch.tensor([k0 * 1.47]))

    def forward(self, xy):
        # Hard-enforce 4-fold symmetry:  $E(x,y) = E(-x,y) = E(x,-y) = E(-x,-y)$ 
        r1 = self.net(xy)
        r2 = self.net(torch.cat([-xy[:, :1], xy[:, 1:]], dim=1))
        r3 = self.net(torch.cat([xy[:, :1], -xy[:, 1:]], dim=1))
        r4 = self.net(torch.cat([-xy[:, :1], -xy[:, 1:]], dim=1))
        return (r1 + r2 + r3 + r4) / 4

def ri_2d(xy):
    inside = (torch.abs(xy[:, 0]) <= w_x / 2) & (torch.abs(xy[:, 1]) <= w_y / 2)
    return torch.where(inside,
                        torch.tensor(n_core, dtype=torch.float32),

```

```
torch.tensor(n_clad, dtype=torch.float32))
    .unsqueeze(1)
```

A.2.3 FDE Solver for Benchmark

```
import scipy.sparse as sp
import scipy.sparse.linalg as spla

print("\nRunning 2D FDE solver for ground truth...")

# Grid
Nx, Ny = 80, 80
x = np.linspace(-dom2d, dom2d, Nx)
y = np.linspace(-dom2d, dom2d, Ny)
dx = x[1] - x[0]
dy = y[1] - y[0]

XX, YY = np.meshgrid(x, y)

# Refractive index map
n_map = np.where((np.abs(XX) <= w_x/2) & (np.abs(YY) <= w_y/2),
                 n_core, n_clad)

# Flatten grid
N = Nx * Ny

def laplacian_2d(Nx, Ny, dx, dy):
    """Construct sparse 2D Laplacian."""
    Ix = sp.eye(Nx)
    Iy = sp.eye(Ny)

    Dx = sp.diags([1, -2, 1], [-1, 0, 1], shape=(Nx, Nx)) / dx**2
    Dy = sp.diags([1, -2, 1], [-1, 0, 1], shape=(Ny, Ny)) / dy**2

    return sp.kron(Iy, Dx) + sp.kron(Dy, Ix)

# Build operator
L = laplacian_2d(Nx, Ny, dx, dy)

n_flat = n_map.flatten()
K = sp.diags((k0**2) * (n_flat**2))

A = L + K

print("Solving sparse eigenvalue problem...")
eigvals, eigvecs = spla.eigs(A, k=1, which='LR')

beta_fde = np.sqrt(np.real(eigvals[0]))
neff_fde = beta_fde / k0
```